

Practical exercises

— interpreting and discussing results

A Cambridge insert: averages, spread, and the graph that tells the truth about the data.

LESSON 166

1 × 40 MIN

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S8 16

LESSON CLOCK

10'

12'

10'

6'

2'

The three averages	10 min
Which average to use	12 min
Comparing two sets	10 min
Misleading graphs	6 min
Homework	2 min

LEARNING OBJECTIVES

- Find the mean, median, mode and range of a small data set.
- Choose the average that describes the data honestly.
- Read a bar chart and a frequency table and compare two sets.
- Recognise a misleading graph.

TEXTBOOK

National	pp. 503–506
Cambridge	Stage 8, pp. 158–166
Workbook	Exercise 16.1

01

Explanation

The three averages

MEASURE	HOW	FOR 3, 5, 5, 6, 8, 9, 12
mean	add and divide by the count	$\frac{48}{7} \approx 6.9$
median	the middle value in order	6
mode	the commonest value	5
range	largest minus smallest	9

ORDER THE DATA BEFORE TAKING A MEDIAN

The middle of the list as written is not the median unless the list is sorted. With an even number of values, average the two middle ones.

Which average to use

DATA	BEST AVERAGE	WHY
test marks, no extremes	mean	it uses every value
salaries with one very large one	median	the outlier drags the mean up
shoe sizes sold	mode	the shop wants the commonest
colours of cars	mode only	they cannot be added or ordered

Example: 2, 3, 3, 4, 88 has mean 20 and median 3. The mean describes no one in the list.

THE RANGE IS NOT AN AVERAGE

It measures spread, not the centre. Two classes can share a mean of 60 with ranges of 10 and 70 — very different classes, and the range is what says so.

Comparing two sets

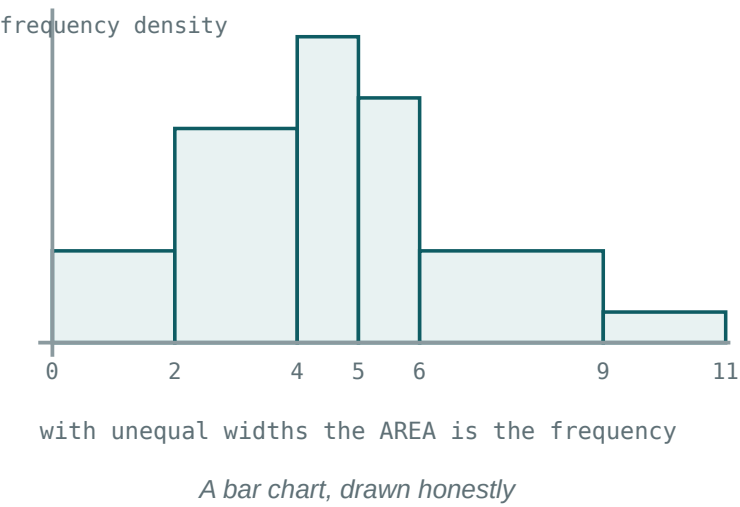
CLASS	MEAN	RANGE	WHAT IT MEANS
7A	62	14	similar marks, tightly grouped
7B	62	55	the same average, wildly spread

A fair comparison quotes both: one measure of centre and one of spread. Quoting the mean alone would make the two classes look identical.

TWO SENTENCES FINISH ANY COMPARISON

“On average 7B did as well as 7A; but 7B’s marks were far more spread out.” Centre, then spread — and both in words, not just numbers.

Misleading graphs



TRICK	WHAT IT DOES	THE FIX
vertical axis not starting at 0	small differences look huge	start at 0, or mark the break
bars of different widths	area misleads the eye	equal widths
no scale on an axis	nothing can be checked	label both axes
pictures scaled in two directions	doubling looks like quadrupling	use bars

ALWAYS READ THE VERTICAL AXIS FIRST

A bar twice as tall as another means twice as much only if the axis starts at zero. This is the commonest way an honest set of numbers is used to tell a lie.

02 Worked examples

EXAMPLE 1 Find the mean, median, mode and range of 3, 5, 5, 6, 8, 9, 12.

- 1 Sum = 48, count 7, so the mean is ≈ 6.9 .
- 2 Already in order; the middle value is 6.
- 3 5 appears twice — it is the mode.
- 4 Range = $12 - 3 = 9$.

ANSWER

Mean ≈ 6.9 , median 6, mode 5, range 9

EXAMPLE 2 Which average best describes 2, 3, 3, 4, 88?

- 1 Mean = $\frac{100}{5} = 20$.
- 2 Median = 3.
- 3 The 88 is an outlier and drags the mean far above every other value.
- 4 The median describes the data honestly.

ANSWER

The median

EXAMPLE 3 Two classes both have mean 62; the ranges are 14 and 55. What can be said?

- ① On average they performed equally.
- ② The second class is far more spread out.
- ③ Some pupils there did much better and some much worse.

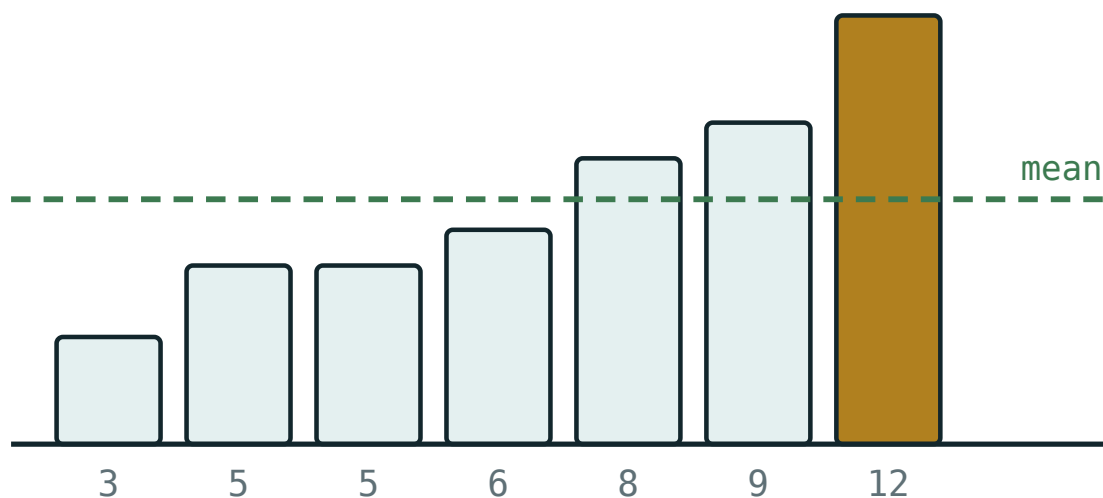
ANSWER

Same centre, very different spread

03 Interactive model

Show the same bar chart twice, once with the axis from 0 and once from 55; the class sees the same data tell two different stories.

Change one value and watch the three averages



values **3, 5, 5, 6, 8, 9, 12**

mean **6.86**

median **6**

mode **5**

range **9**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Mean	O'rta arifmetik	Среднее арифметическое
Median	Mediana	Медиана
Mode	Moda	Мода
Range	Ayirma (tarqoqlik)	Размах
Outlier	Chetlangan qiymat	Выброс
Bar chart	Ustunli diagramma	Столбчатая диаграмма
To compare	Taqqoslash	Сравнить
Misleading	Chalg'ituvchi	Вводящий в заблуждение

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

The mean of 3, 5, 5, 6, 8, 9, 12 is about:

Its median is:

Its mode is:

Its range is:

3

6

9

12

With one huge outlier, the best average is the:

mean

median

mode

range

A bar chart whose axis starts at 55 is:

fine

misleading

impossible

a histogram

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. The mean of 4, 6, 8

ANSWER 6

2. The median of 4, 6, 8

ANSWER 6

3. The mode of 4, 6, 6, 8

ANSWER 6

4. The range of 4, 6, 8

ANSWER 4

5. The mean of 3, 5, 5, 6, 8, 9, 12

ANSWER ≈ 6.9

6. Its median

ANSWER 6

7. Its range

ANSWER 9

Medium · the standard question

7 PROBLEMS

1. The mean of 2, 3, 3, 4, 88

ANSWER 20

2. Its median

ANSWER 3

3. Which average describes it better?

ANSWER The median

4. The median of 4, 7, 9, 12

ANSWER 8

5. Two classes, mean 62, ranges 14 and 55

ANSWER Same centre, different spread

6. Which average suits shoe sizes sold?

ANSWER The mode

7. Which suits car colours?

ANSWER The mode only

Hard · stretch and reason

7 PROBLEMS

1. Five numbers have mean 10: their sum

ANSWER 50

2. Four numbers have mean 7; a fifth is added and the mean becomes 8: the fifth number

ANSWER 12

3. A set of 5 numbers with mean 6, median 5 and mode 5

ANSWER e.g. 3, 5, 5, 7, 10

4. Why can the mean lie outside the commonest values?

ANSWER One extreme value pulls it away

5. Two sets with the same mean but ranges 2 and 40

ANSWER One is consistent, the other erratic

6. Why must a bar chart axis start at 0?

ANSWER Otherwise bar heights are not proportional

7. The mean of the first 10 whole numbers 1 to 10

ANSWER 5.5

07 Homework

Homework — 5 tasks

Order the data first; the median depends on it and the range does too.

- Find the mean, median, mode and range of 7, 4, 9, 4, 6, 10, 4.
- For 1, 2, 2, 3, 92, find the mean and median and say which describes the data better.
- Two classes have mean 55 with ranges 12 and 48. Write two sentences comparing them.
- Draw a bar chart of your own data with the axis starting at 0.
- Describe two ways a graph can mislead.